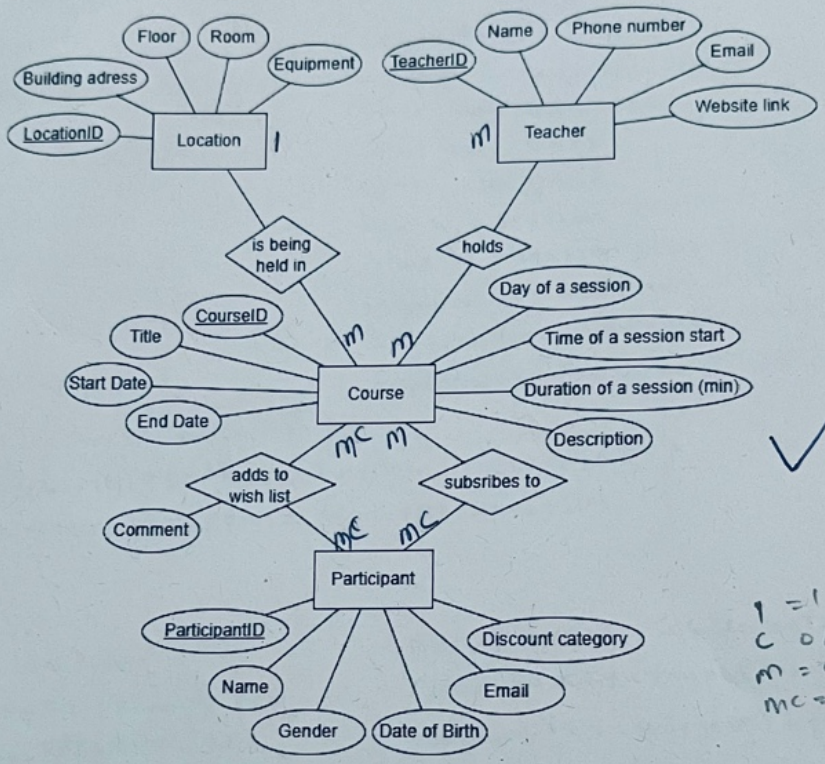
## Task 1 Data Modeling (20 points)




(a) Given the above Entity-Relationship diagram, specify the cardinalities in Modified Chen notation based on the following information. **(8 points)**

- The platform connects people teaching at different locations with people who want to participate in their courses.
- A person is only listed on the platform as a teacher if they provide at least one course through it. It is also possible for multiple people to co-teach the same course simultaneously. But there cannot be a course without a teacher who supervises it.
- A course takes place only at one location assigned to it throughout its duration. However, in a week multiple courses can take place in the same location as long as there are no scheduling conflicts. A location is only listed on the platform if at least one course takes place in it.
- A person is only listed on the platform as a participant if they subscribe to at least one course through it. A course can have an arbitrary number of participants.
- A participant can also add other courses to which they are not subscribed to a wish list. There is no limit to the number of courses in a single participant. A course can be added to an arbitrary number of participants' wish lists.

**(b)** Map the given Entity-Relationship diagram into a relational schema, including data types, primary keys, and foreign keys. **(12 points)** 10,75

```
CREATE TABLE Teacher(
TeacherID INTEGER PRIMARY KEY,
Name VARCHAR(128),
phonenumber ~~INTEGER~~ duchem,
Email VARCHAR(128),
Website link VARCHAR
);

CREATE TABLE holds(
PRIMARY KEY(TeacherID, CourseID),
FOREIGN KEY (TeacherID) REFERENCES Teacher(TeacherID),
FOREIGN KEY (courseID) REFERENCES Course(CourseID)
);

CREATE TABLE location(
LocationID INTEGER PRIMARY KEY,
Building address VARCHAR(128),
Floor ~~VARCHAR~~ (128),
Room VARCHAR(128),
Equipment VARCHAR(128),
);

CREATE TABLE Participant(
ParticipantID INTEGER PRIMARY KEY,
Name VARCHAR(128),
Gender VARCHAR(128),
Date of Birth DATE,
Email VARCHAR(128),
Discount category VARCHAR(128),
);

CREATE TABLE Course(
CourseID INTEGER PRIMARY KEY,
Title VARCHAR(128),
Start Date DATE,
End Date DATE,
Day of a session varchar - 0,25
Time of a session start time - 0,25
Duration of a session int - 0,25
Description VARCHAR(128),
FOREIGN KEY (LocationID) REFERENCES location
(LocationID)
);

CREATE TABLE Subscribes to (
PRIMARY KEY (CourseID, ParticipantID),
FOREIGN KEY (CourseID) REFERENCES Course
(CourseID),
FOREIGN KEY (participant ID) REFERENCES
participant(participantID)
);

CREATE TABLE adds to wishlist (
PRIMARY KEY (CourseID, participantID),
FOREIGN KEY(CourseID) REFERENCES
Course(CourseID),
FOREIGN KEY (ParticipantID) REFERENCES
Participant(participantID)
); comment - 0,5
```

10

## Task 2 Structured Query Language (30 points)

| CourseID | Title | Start_Date | End_Date | Day_of_session | Time_of_session_start | Duration_of_session | Description | LocationID |
|---|---|---|---|---|---|---|---|---|
| 1 | Soccer Advanced | 2025-03-03 | 2025-06-30 | Monday | 10:00:00 | 90 | Soccer techniques for experienced players. | 1 |
| 2 | Swimming for Beginners | 2025-03-05 | 2025-06-25 | Wednesday | 14:00:00 | 90 | Basic strokes and water safety | 3 |
| 3 | Tennis Techniques | 2025-03-04 | 2025-04-29 | Tuesday | 13:00:00 | 60 | Serve, volley, footwork | 2 |
| 4 | Volleyball Drills | 2025-03-06 | 2025-04-24 | Thursday | 15:00:00 | 60 | Passing, setting, spiking | 5 |
| 5 | Yoga for Athletes | 2025-03-04 | 2025-06-24 | Tuesday | 08:00:00 | 90 | Flexibility and strength | 9 |
| 7 | Table Tennis Advanced | 2025-03-06 | 2025-06-26 | Thursday | 16:00:00 | 120 | Spins, smashes, rallies | 11 |
| 8 | Golf for Beginners | 2025-03-01 | 2025-06-28 | Saturday | 10:00:00 | 120 | Swing, putting, etiquette | 8 |
| 9 | Canoeing | 2025-03-07 | 2025-06-27 | Friday | 11:00:00 | 120 | Paddling and navigation | 6 |
| 10 | Kayaking | 2025-03-07 | 2025-06-27 | Friday | 13:00:00 | 120 | Paddling and navigation | 6 |

| TeacherID | Name | Phone_number | Email | Website_link |
|---|---|---|---|---|
| 1 | Alice Johnson | 555-123-4567 | alice.johnson@yo.com | https://www.linkedin.com/in/alicej |
| 2 | Bob Smith | 555-234-5678 | bob.smith@yo.com | http://bobsmith.org |
| 3 | Carla Nguyen | 555-345-6789 | carla.nguyen@yo.com | https://facebook.com/carla.nguyen |
| 4 | David Lee | 555-456-7890 | david.lee@yo.com | http://davidlee.me |
| 5 | Eva Brown | 555-567-8901 | eva.brown@yo.com | https://www.linkedin.com/in/evabrown |
| 6 | Frank White | 555-678-9012 | frank.white@yo.com | http://frankwhite.info |
| 7 | Grace Kim | 555-789-0123 | grace.kim@yo.com | http://gracekim.org |
| 8 | Henry Thompson | 555-890-1234 | henry.thompson@yo.com | https://facebook.com/henrythompson |
| 9 | Liam Moore | 555-222-3333 | liam.moore@yo.com | http://liammoore.com |
| 10 | Quinn Brooks | 555-777-8888 | quinn.brooks@yo.com | https://www.linkedin.com/in/qbrooks |

| SupervisesID | CourseID | TeacherID |
|---|---|---|
| 1 | 2 | 7 |
| 2 | 3 | 1 |
| 3 | 4 | 10 |
| 4 | 9 | 4 |
| 5 | 7 | 6 |
| 6 | 10 | 4 |
| 7 | 10 | 3 |
| 8 | 5 | 2 |
| 9 | 8 | 9 |
| 10 | 9 | 3 |

(a) Given the Course, Teacher and CourseTeacher tables above, compute the results for the following three queries: **(15 points - 5 per query)**

**[Q1]**
```
SELECT Title, End_Date
FROM Course
WHERE End_Date < '2025-05-01';
```

Title | End_Date
4 VD
3 17

5

**[Q2]**
```
SELECT Day_of_session, AVG(Duration_of_session) as Avg_Duration
FROM Course
GROUP BY Day_of_session
ORDER BY Avg_Duration DESC;
```

Mon 90
Wed 90
Tu 75
Thur 90
Sat 120
Fri 120

5

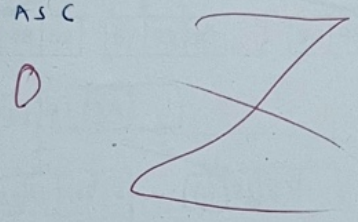
**[Q3]**
```
SELECT t.Name, COUNT(ct.CourseID) as Course_Count
FROM Teacher t
JOIN CourseTeacher ct ON t.TeacherID = ct.TeacherID
GROUP BY t.TeacherID
HAVING COUNT(ct.CourseID) > 1;
```

0

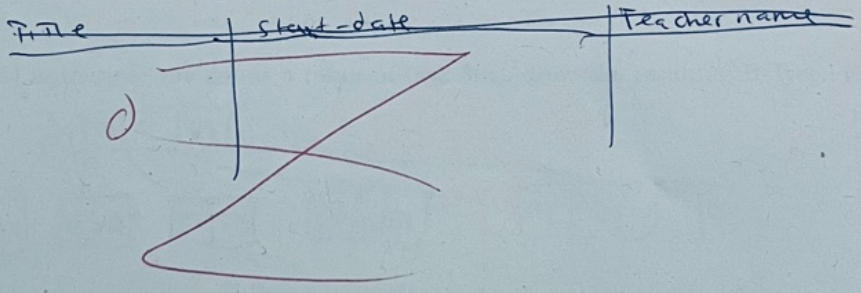
**(b)** Given all the tables of the dataset from the task 1, write SQL queries to answer the following questions (in a way that is independent of the data shown in task 2(a)): **(15 points - 5 per query)**

[Q4] Find all pairs of courses that take place on the same day and in the same location (return title of one course, title of another one, day and location ID).

```
Select Title.C, Day-of-session, location ID
FROM Courses AS C
```

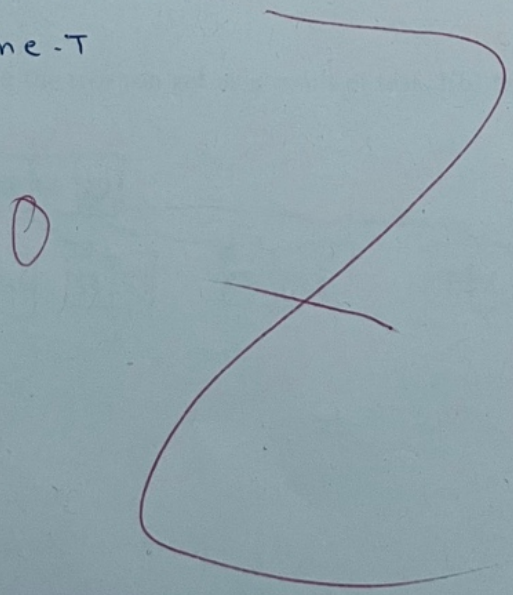


[Q5] Get the earliest starting course(s) and its teacher(s) in terms of starting date (return title of the course, starting date and teacher name).



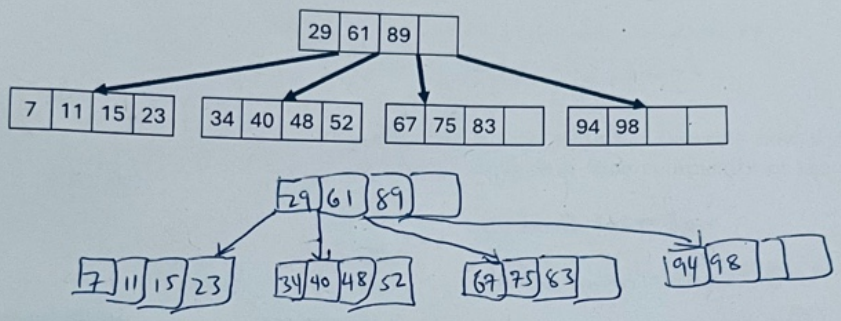

[Q6] List teachers who supervise only one course and that course lasts more than 90 minutes (return teacher name).

```
Select Name.T
```



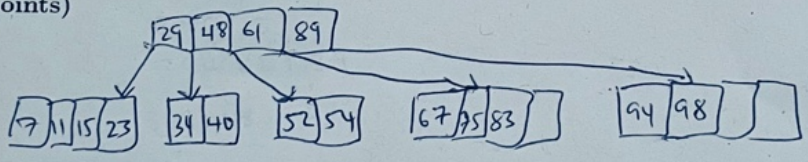
## Task 3 Physical Design (6 points)

(a) Given the B-tree (k=2) below, find an error in it and fix it, draw the resulting B-Tree. **(2 points)**




2

(b) Insert key 54 in the tree you got as a result of task 3(a), draw the resulting B-Tree. **(2 points)**




2

(c) Delete key 89 from the tree you got as a result of task 3(b) and draw the final B-tree. **(2 points)**




2

## Task 4 Relational Algebra and Query Processing (5 points)

(a) What SQL keyword does this relational algebra operator correspond to: $\pi_{i1,\ldots,im}(R)$? What purpose does it serve in set and bag semantics? **(1 point)**

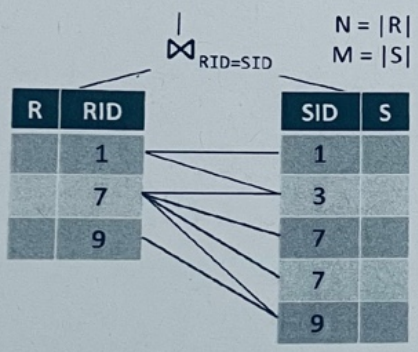
1

The Select Statement

Set semantics - Distinct or Unique elements

Bag semantics - Duplicates are considered

(b) What type of operator is represented by this picture? Describe how it functions (walk us through its stages). What are the space and time complexity of this operation? **(4 points)**




1

Nested loop Join

Space complexity :- O(N * M) ✗

Time complexity :- O(1) ✓

## Task 5 Impedance Mismatch (4 points)

Explain the concept of impedance mismatch in database systems. Why does it occur, and what is a common strategy to resolve this issue?

0.5 Impedance mismatch strategy - ORM